Homework 1

Incentive-Centered Design

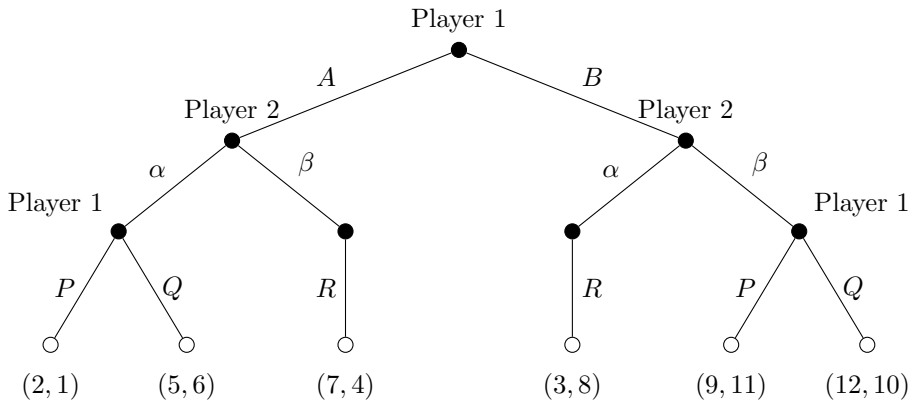
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2. LLMs referred: ChatGPT, Claude

Solutions

1. Consider a 2-player extensive form game specified by the following Tree. At time $t = 1$ and $t = 3$, it is Player 1's turn to play, and at time $t = 2$, it is Player 2's turn to play. The available actions at each time are specified in the Tree, and both players know the Tree.



   (a) What action do you think Player 1 will play at time 1, Player 2 will play at time 2, and Player 3 will play at time 3?

   (b) Identify the set of strategies available to each player.

   (c) Express the game in strategic form.

   (d) Identify all pure-strategy Nash equilibria of the game.

   (e) Is the action sequence identified in Part (a) generated by any Nash equilibrium? Are any other action sequences also generated by a Nash equilibrium?

   (f) In the presence of a multiplicity of Nash equilibria, refined notions of equilibria can be defined to weed out certain Nash equilibria. One such notion for extensive-form games is *subgame-perfect equilibrium.*

   A subgame of an extensive-form game is a game defined by any sub-Tree of the game. For example, in the above game, one can begin observing the game from $t = 2$ so that the first move is played by Player 2 and the second move is played by Player 1. There are two such 2-round sub-games in the above game — one corresponding to each node in the second layer. Similarly, there are four 1-round games. The above game, therefore, has seven sub-games (the original game is also a subgame).

   A Nash equilibrium of an extensive-form game is said to be a subgame-perfect equilibrium if the strategy profile, when implemented on any subgame, also forms a Nash equilibrium. Read up more on subgame-perfect equilibrium from your favorite source.

   Which of the Nash equilibria identified in Part (d) are subgame-perfect equilibria? Do the action sequences generated by subgame-perfect equilibria coincide with the action sequence you identified in Part (a)?

Solution:

(a) at time 1, player 1 will pay strategy B because the gains on the right branch of the tree for player 1 is always higher than the gains in the left branch of the tree.
At time 2, player 2 plays while also considering the fact that player 1 will play strategy B in time 1, in the right branch, player 2 has better gains if he chooses to play strategy β.
At time 3, Player 1 will account all the previous moves and prefer path Q since he/she would have a better gain. Therefore, at time 3 player 1 will play Q.

(b) We define the action set available to player 1 as:

$$S_{1\{t=1\}} = \{A, B\} \qquad S_{1\{t=3\}} = \{P, Q, R\}$$

but the R node is forced, it is never a choice, we can therefore write
- At node 1 : A or B
- After A-α : P or Q
- After B-β : P or Q

Hence, player 1 has 8 strategies :

$$S_1 = \{(A,P,P),(A,P,Q),(A,Q,P),(A,Q,Q),(B,P,P),(B,P,Q),(B,Q,P),(B,Q,Q)\}$$

We define the action set available to player 2 as:

$$S_{2\{t=2\}} = \{\alpha, \beta\}$$

- After A : α or β
- After B : α or β

$$S_2 = \{\alpha\alpha, \alpha\beta, \beta\alpha, \beta\beta\}$$

(c) We can club player 1's strategy and player 2's strategy and normalize the extensive game to a standard form as follows.

	Player 1							
Player 2	(A,P,P)	(A,P,Q)	(A,Q,P)	(A,Q,Q)	(B,P,P)	(B,P,Q)	(B,Q,P)	(B,Q,Q)
(α,α)	$(2,1)$	$(2,1)$	$(5,6)$	$(5,6)$	$(3,8)$	$(3,8)$	$(3,8)$	$(3,8)$
(α,β)	$(2,1)$	$(2,1)$	$(5,6)$	$(5,6)$	$(9,11)$	$(12,10)$	$(9,11)$	$(12,10)$
(β,α)	$(7,4)$	$(7,4)$	$(7,4)$	$(7,4)$	$(3,8)$	$(3,8)$	$(3,8)$	$(3,8)$
(β,β)	$(7,4)$	$(7,4)$	$(7,4)$	$(7,4)$	$(9,11)$	$(12,10)$	$(9,11)$	$(12,10)$

(d) The nash equilibria are the following:
- (B, P, Q) x (α,β)
- (B, Q, Q) x (α,β)
- (B, P, Q) x (β,β)
- (B, Q, Q) x (β,β)

(e) Yeah action sequence (a) is generated by the nash equilibrium, no other action sequences are not generated by the nash equilibrium.

(f) Analysis of sub games:
- After $A-\alpha$: Player 1 will choose Q since he has a pay off of 5
- After $B-\beta$: Player 1 will choose Q since he has a pay off of 12
- After A : Player 2 will choose α since he has better gains
- After B : Player 2 will choose β since he has better gains
- at node 1: Player 1 will choose B since he has higher pay off

Therefore the sub-game perfect equilibrium converges to (B, Q, Q) and (α,β). Yes it generates the action sequence in part (a).

2. This question asks you to prove the proposition in Lecture 4, Slide 15. Let $\mathcal{X} \subseteq \mathbb{R}^m$ and $\mathcal{Y} \subseteq \mathbb{R}^n$ be any two sets, and let $f : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ be any function. Show that there exists $(\mathbf{x}^*, \mathbf{y}^*)$ satisfying

$$\mathbf{x}^* = \arg\max_{\mathbf{q}} \quad f(\mathbf{q}, \mathbf{y}^*), \tag{1}$$

$$\mathbf{y}^* = \arg\min_{\mathbf{q}} \quad f(\mathbf{x}^*, \mathbf{q}) \tag{2}$$

if and only if

$$\max_x \min_y f(x, y) = \min_y \max_x f(x, y). \tag{3}$$

Solution:

Forward Proof

Let there exist a $(\mathbf{x}^*, \mathbf{y}^*)$ satisfying

$$\mathbf{x}^* = \arg\max_{\mathbf{q}} \quad f(\mathbf{q}, \mathbf{y}^*), \tag{4}$$

$$\mathbf{y}^* = \arg\min_{\mathbf{q}} \quad f(\mathbf{x}^*, \mathbf{q}) \tag{5}$$

$$f(x, y^*) \leq f(x^*, y^*) \quad \forall x \quad and \quad f(x^*, y) \geq f(x^*, y^*) \quad \forall y$$

$$\implies f(x^*, y^*) = \min_y f(x^*, y)$$

$$\max_x \min_y f(x, y) \geq \min_y f(x^*, y) = f(x^*, y^*)$$

Similarly,

$$\implies f(x^*, y^*) = \max_x f(x, y^*)$$

$$\min_y \max_x f(x, y) \leq \max_x f(x, y^*) = f(x^*, y^*)$$

$$\implies \min_y \max_x f(x, y) \leq \max_x \min_y f(x, y) \tag{6}$$

We already have this standard result (best of worst cases is sub-optimal to worst of best cases):

$$\min_y f(x, y) \leq f(x, y) \leq \max_x f(x, y)$$

$$\Rightarrow \min_y f(x, y) \leq \max_x f(x, y)$$

$$\Rightarrow \max_x \min_y f(x, y) \leq \min_y \max_x f(x, y) \tag{7}$$

Combining equations (6) and (7) we conclude that,

$$\max_x \min_y f(x, y) = \min_y \max_x f(x, y)$$

Backward Proof

$$\text{Assume } \max_x \min_y f(x, y) = \min_y \max_x f(x, y) = v$$

$$\exists\, x^* \text{ such that } \min_y f(x^*, y) = v, \quad \exists\, y^* \text{ such that } \max_x f(x, y^*) = v$$

$$\Rightarrow \forall y : \ f(x^*, y) \geq v, \quad \forall x : \ f(x, y^*) \leq v$$

$$\Rightarrow f(x, y^*) \leq f(x^*, y^*) \leq f(x^*, y)$$

$$\Rightarrow x^* \in \arg\max_x f(x, y^*), \quad y^* \in \arg\min_y f(x^*, y)$$

Hence proved.

3. Consider a two-player zero-sum game with the maximizer having m pure strategies and the minimizer having n pure strategies. Denote by $\mathbf{x} \in \mathbb{R}^m$ the mixed strategy of the maximizer, by $\mathbf{y} \in \mathbb{R}^n$ the mixed strategy of the minimizer, and by $\mu(\mathbf{x}, \mathbf{y})$ the expected payoff of the maximizer. Show that $\mu : \mathbb{R}^m \times \mathbb{R}^n \to \mathbb{R}$ is concave-convex, i.e., $\mu(\cdot, \mathbf{y})$ is concave for each $\mathbf{y}$ and $\mu(\mathbf{x}, \cdot)$ is convex for each $\mathbf{x}$.

Solution:

We can define the expected payoff of the maximizer as,

$$\mu(x, y) = x^T A y$$

where, A is the payoff matrix of the maximizer.

Part A: concave

Let,

$$f(x) = \mu(\cdot, \mathbf{y}) = x^T A y$$
$$\nabla f(x) = Ay \qquad \nabla^2 f(x) = 0$$

Hence, this passes the concavity test in second derivative $f"(x) \leq 0$. Therefore, $\mu(\cdot, \mathbf{y})$ is concave for each y .

Part B: Convex

Let,

$$f(y) = \mu(\mathbf{x}, \cdot) = x^T A y$$
$$\nabla f(y) = x^T A \qquad \nabla^2 f(y) = 0$$

Hence, this passes the convexity test in second derivative $f"(y) \geq 0$. Therefore, $\mu(\mathbf{x}, \cdot)$ is convex for each x.

Hence proved

4. Consider a 2-player normal form game in which the row player has m pure strategies and the column player has n pure strategies. Suppose that no two strategy profiles have the same payoff for any player. What is maximum number of Pure-strategy Nash Equilibria (PNE) that the game can have?

   If not for the constraint that all payoffs are distinct, what is the maximum number of PNE that the game can have?

   Solution:

   Lets assume that our payoff matrix is A $(m \times n)$. To identify the nash equilibrium we compute the best row player strategy for each column and the best column player strategy for each row. Since no two strategy profiles have the same payoff for any player,
   We will identify n row player strategies and m column player strategies, now we identify the nash equilibrium by computing the overlap. When we maximize that overlap we get,

$$\#PNE = \min(m, n)$$

   If the payoffs are not distinct, then we can have multiple strategy profiles that are optimal, to maximize the equilibrium we make every profile best profile (i.e, the payoff matrix is constant). In which case,

$$\#PNE = m \times n$$

5. Identify all pure-strategy Nash equilibria and mixed-strategy Nash equilibria of the game with the payoff matrix

$$\begin{bmatrix} (1,2) & (1,3) \\ (0,3) & (4,1) \end{bmatrix}. \tag{8}$$

Solution:

Pure strategy Nash Equilibria :

For row 1: (1,3) is the best payoff for the column player
For row 2: (0,3) is the best payoff for the column player
For col 1: (1,2) is the best payoff for the row player
For col 2: (4,1) is the best payoff for the row player

Therefore a **pure strategy nash equilibria doesn't exist**.

Mixed Strategy Nash Equilibria :

Assume that $\alpha, 1-\alpha$ and $\beta, 1-\beta$ be the mixed stretegy profiles for player 1 and 2.

If player 1 is indifferent, then expected payoffs from row 1 and row 2 are equal,

$$row_1 = \beta + 1 - \beta = 1 \qquad row_2 = 4(1-\beta)$$

$$\implies \beta = \frac{3}{4}$$

Similarly, for nash equilibrum, the exepcted payoffs from col 1 and col 2 are equal for player 2,

$$col_1 = 3 - \alpha \qquad 1 + 2\alpha$$

$$\implies \alpha = \frac{2}{3}$$

Theerfore the mixed strategy nash equilibrium is given by $(\frac{2}{3}, \frac{3}{4})$.

6. A potential game consisting of n players is a normal form game with the property that there exists a function Φ, called a potential function, such that for every $i \in \{1, \ldots, n\}$, every $s_i, s_i' \in \mathcal{S}_i$, and every $\mathbf{s}_{-i} \in \mathcal{S}_{-i}$,

$$u_i(s_i, \mathbf{s}_{-i}) - u_i(s_i', \mathbf{s}_{-i}) = \Phi(s_i, \mathbf{s}_{-i}) - \Phi(s_i', \mathbf{s}_{-i}) \tag{9}$$

where u_i is the payoff function of player i. Prove that all potential games have a pure-strategy Nash equilibrium.

Remark: Potential games derive their importance from the fact that all *congestion games* have a potential function associated with them. Congestion games model situations wherein a multitude of self-interested agents share common resources. Examples include transportation networks where vehicles share the roads, communication networks where packets share the links, cloud computing where users share clock cycles of the CPU, retail warehousing where sellers share space and manpower of the retail outlets, etc. The above proof implies that every congestion game has a Nash equilibrium in pure strategies.

Solution:

Since the strategy space is finite, there exists a strategy profile

$$\mathbf{s}^* \in \arg\max_{\mathbf{s}} \Phi(\mathbf{s}).$$

We need to prove that $\mathbf{s}^*$ is a pure-strategy Nash equilibrium.

For any player i, consider any unilateral deviation $s_i' \in \mathcal{S}_i$. Then, by the definition of a potential game,

$$u_i(s_i', \mathbf{s}_{-i}^*) - u_i(s_i^*, \mathbf{s}_{-i}^*) = \Phi(s_i', \mathbf{s}_{-i}^*) - \Phi(s_i^*, \mathbf{s}_{-i}^*).$$

Since $\mathbf{s}^*$ maximizes Φ, we have:

$$\Phi(s_i', \mathbf{s}_{-i}^*) - \Phi(s_i^*, \mathbf{s}_{-i}^*) \leq 0.$$

Thus,

$$u_i(s_i', \mathbf{s}_{-i}^*) - u_i(s_i^*, \mathbf{s}_{-i}^*) \leq 0,$$

which implies

$$u_i(s_i^*, \mathbf{s}_{-i}^*) \geq u_i(s_i', \mathbf{s}_{-i}^*) \quad \forall s_i' \in \mathcal{S}_i.$$

This proves that for any player i, the payoff is maximized at the optimal strategy profile $\mathbf{s}^*$ where,

$$\mathbf{s}^* \in \arg\max_{\mathbf{s}} \Phi(\mathbf{s}).$$

Since, the potential function $\Phi(.)$ can be maximized, we have a nash equilibrium for potential game which is given by $\mathbf{s}^*$.

7. Construct an example and a non-example of a 2-player potential game wherein each player has 4 actions. Prove that the games you construct are indeed potential and non-potential games respectively.

Solution:

An example of a two player potential game is as follows:

$$A = \begin{bmatrix} (2,2) & (0,0) \\ (0,0) & (1,1) \end{bmatrix}$$

Define the potential function:

$$\Phi = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

Since $u_1 = u_2 = \Phi$, any unilateral deviation by a player yields:

$$u_i(s_i', s_{-i}) - u_i(s_i, s_{-i}) = \Phi(s_i', s_{-i}) - \Phi(s_i, s_{-i})$$

Hence, this is a potential game.

An example of a non-potential game is given as follows

$$A = \begin{bmatrix} (1,0) & (0,1) \\ (0,1) & (1,0) \end{bmatrix}$$

Check deviation from from row 1 to row 2 in column 1:

$$u_1(2,1) - u_1(1,1) = 0 - 1 = -1$$
$$u_2(2,1) - u_2(1,1) = 1 - 0 = 1$$

The payoff changes differ, so they cannot be represented by a single function Φ.

8. This question pertains to the comments in Lecture 6, Slide 6. Consider a two-player zero-sum game and denote by $f(\mathbf{x}, \mathbf{y})$ the expected payoff of the maximizer when the maximizer employs the mixed strategy $\mathbf{x}$ and the minimizer employs the mixed strategy $\mathbf{y}$. Let $(\mathbf{x}^*, \mathbf{y}^*)$ be a Nash equilibrium of the game.

   (a) Show that by employing the strategy $\mathbf{x}^*$, the maximizer is assured an expected payoff of at least $f(\mathbf{x}^*, \mathbf{y}^*)$ regardless of the strategy that the minimizer employs.

   (b) Show that by employing the strategy $\mathbf{y}^*$, the minimizer can keep the maximizer from obtaining an expected payoff of more than $f(\mathbf{x}^*, \mathbf{y}^*)$ regardless of the strategy that the maximizer employs.

   Remark: Blackwell Approachability — a celebrated idea in Game theory with applications in many other fields — is a generalization of the above result for vector-valued payoffs.

   Solution:

   (a) At equilibrium the minimizer tries to minimize the payoff of the maximizer. Hence, the mixed strategy profile minimizes y^* the payoff function.

$$f(x^*, y^*) = \min_y f(x^*, y)$$

$$\implies f(x^*, y^*) \leq f(x^*, y) \qquad \forall y$$

   Thus by employing a mixed strategy profile x^* the maximizer ensures an expected payoff of atleast $f(x^*, y^*)$ irrespective of the strategy profile the minimizer implements.

   (b) At equilibrium the maximizer tries to maximize his/her payoff function. Hence the mixed strategy profile x^* maximizes the payoff function.

$$f(x^*, y^*) = \max_x f(x^*, y)$$

$$\implies f(x^*, y^*) \geq f(x, y^*) \qquad \forall x$$

   Thus by employing a mixed strategy profile y^* the minimizer ensures that the maximizer doesn't obtain an expected payoff more than $f(x^*, y^*)$ irrespective of the strategy profile the maximizer implements.

   Hence proved.

9. Compute the Nash equilibrium of the two-player zero-sum game with the maximizer's payoff matrix

$$\begin{bmatrix} 2 & -4 & 7 \\ -1 & 3 & -4 \\ 5 & -8 & 2 \\ -4 & 5 & 1 \end{bmatrix}. \tag{10}$$

Compute the value of the game.

You can use Matlab (or any other programming language of your choice) to solve Linear Programs.

Solution:

The payoff function of the maximizer is given by,

$$f(x, y) = x^T A y$$

where x and y are the mixed strategy profiles of the maximizer and minimizer respectively. And A is given by,

$$A = \begin{bmatrix} 2 & -4 & 7 \\ -1 & 3 & -4 \\ 5 & -8 & 2 \\ -4 & 5 & 1 \end{bmatrix}.$$

We need to maximize the payoff function of the maximizer. The optimization problem can be written as the following LP,

$$\begin{aligned} \max_{t,x} \quad & t \\ \text{s.t.} \quad & x^T A y \geq t \quad \forall\, y \in \Delta(S_2), \\ & \sum_{j=1}^{m} x_j = 1, \\ & 0 \leq x_j \leq 1, \quad j = 1, \ldots, m. \end{aligned}$$

We can reduce the LP to a finite number of equations for each pure strategy of the minimizer e_i as follows

$$\begin{aligned} \max_{t,x} \quad & t \\ \text{s.t.} \quad & x^T A e_i \geq t \qquad i = 1, \ldots, n. \\ & \sum_{j=1}^{m} x_j = 1, \\ & 0 \leq x_j \leq 1, \quad j = 1, \ldots, m. \end{aligned}$$

The linear program was solved using the implementation available at this link. The resulting optimal strategies and the value of the game are given below:

$$\text{Maximizer's optimal strategy: } x^* = (0.375,\ 0.61029412,\ 0.01470588,\ 0)$$

$$\text{Minimizer's optimal strategy: } y^* = (0.58088235,\ 0.35294118,\ 0.06617647)$$

$$\text{Value of the game: } v = 0.21323529411764724$$

10. The proof of existence of Nash equilibrium that we saw in class has a couple of minor gaps, which you will fill in this question. The notation used in this question is the same as in the lecture slides.

    (a) Show rigorously that the set $\mathcal{X}$ is nonempty, compact, and convex.

    (b) Show that the expected utility function μ_i of any player i is continuous. Where in the proof is the continuity of μ_is used?

    (c) Let $\mathcal{P}_j \subseteq \mathbb{R}^{m_j}$, $j = 1, \ldots, n$, be convex sets. Show that $\mathcal{P}_1 \times \ldots \times \mathcal{P}_n$ is a convex subset of $\mathbb{R}^{m_1} \times \ldots \times \mathbb{R}^{m_n}$. Where in the proof do we use this fact?

Solution:

(a) We know that $\mathcal{X}$ denotes the set of the mixed strategy profiles of all the players

$$\mathcal{X} = \Delta(S_1) \times \cdots \times \Delta(S_n)$$

Since, each player has a non-zero mixed strategy profile $\Delta(S_i)$ is non-empty, the cartesian product of non-empty sets is also nonempty.

$$\Delta(S_i) = \left\{ x_i \in \mathbb{R}^{|S_i|} \;\middle|\; x_i \geq 0,\ \sum_{s_i \in S_i} x_i(s_i) = 1 \right\}$$

Clearly, $\Delta(S_i)$ is bounded since all it's elements lie between 0 and 1.
Let $\{x_i^{(k)}\}_{k=1}^{\infty}$ be a sequence in $\Delta(S_i)$ such that $x_i^{(k)} \to x_i \in \mathbb{R}^{|S_i|}$.
We will show that $x_i \in \Delta(S_i)$.
Since $x_i^{(k)} \in \Delta(S_i)$ for all k, we have:

$$x_i^{(k)}(s) \geq 0 \quad \forall s \in S_i,\ \forall k,$$

and

$$\sum_{s \in S_i} x_i^{(k)}(s) = 1 \quad \forall k.$$

Taking limits coordinate-wise (since convergence in $\mathbb{R}^{|S_i|}$ is coordinate-wise), we get:

$$x_i(s) = \lim_{k \to \infty} x_i^{(k)}(s) \geq 0 \quad \forall s \in S_i.$$

Also, using continuity of finite sums,

$$\sum_{s \in S_i} x_i(s) = \sum_{s \in S_i} \lim_{k \to \infty} x_i^{(k)}(s) = \lim_{k \to \infty} \sum_{s \in S_i} x_i^{(k)}(s) = \lim_{k \to \infty} 1 = 1.$$

Thus, $x_i \in \Delta(S_i)$.
Hence, $\Delta(S_i)$ contains all its limit points and is therefore closed.
Let $x_i, y_i \in \Delta(S_i)$ and let $\lambda \in [0, 1]$. Define

$$z_i = \lambda x_i + (1 - \lambda) y_i.$$

$$\textit{Since}\ ,\ \ x_i \geq 0 \quad ; \quad y_i \geq 0 \implies z_i \geq 0$$

$$\sum_i z_i = \sum_i \lambda x_i + (1 - \lambda) y_i = \lambda \sum_i x_i + (1 - \lambda) \sum_i y_i = \lambda + 1 - \lambda = 1$$

$$\implies z_i \in \Delta(S_i)$$

Hence, convexity is proved.

(b) Each coordinate function $x_j(s_j)$ is continuous on $\mathbb{R}^{|S_j|}$. Hence, for any fixed $(s_1, \ldots, s_n)$, the function

$$(x_1, \ldots, x_n) \mapsto x_1(s_1) \cdots x_n(s_n)$$

is continuous as a product of continuous functions.
Since $u_i(s_1, \ldots, s_n)$ is a constant, each term

$$u_i(s_1, \ldots, s_n)\, x_1(s_1) \cdots x_n(s_n)$$

is continuous. The function μ_i is a finite sum of such terms, hence continuous.
In the proof, we use this property to explain why we have an optimal (x^*, y^*).
Fix x_{-i}. Then $\mu_i(x_i, x_{-i})$ is continuous in x_i. Since $\Delta(S_i)$ is compact, $\mu_i(\cdot, x_{-i})$ attains its maximum on $\Delta(S_i)$. Hence, a best response exists. (Also required for Kakutani's fixed point theorem)

(c) Let $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_n)$ be elements of $\mathcal{P}_1 \times \cdots \times \mathcal{P}_n$, where $x_j, y_j \in \mathcal{P}_j$ for each j.
Let $\lambda \in [0, 1]$ and define

$$z = \lambda x + (1 - \lambda)y = (\lambda x_1 + (1 - \lambda)y_1, \ldots, \lambda x_n + (1 - \lambda)y_n).$$

Since each $\mathcal{P}_j$ is convex,

$$\lambda x_j + (1 - \lambda)y_j \in \mathcal{P}_j \quad \text{for all } j.$$

Thus, $z \in \mathcal{P}_1 \times \cdots \times \mathcal{P}_n$, proving convexity.
This fact is used to show that $\mathcal{X} = \Delta(S_1) \times \cdots \times \Delta(S_n)$ is convex, since each $\Delta(S_i)$ is convex (part a). (Also required for Kakutani's fixed point theorem)

11. Prove that every Nash Equilibrium is a Correlated Equilibrium, and that every Correlated Equilibrium is a Coarse Correlated Equilibrium.

Solution:

(1) Every Nash Equilibrium is a Correlated Equilibrium

Let $\sigma = (\sigma_1, \ldots, \sigma_n)$ be a (mixed) Nash equilibrium. Define a distribution π over action profiles by taking the product distribution:

$$\pi(a) = \prod_{i=1}^{n} \sigma_i(a_i).$$

We show that π satisfies the conditions of a correlated equilibrium. For any player i, and any actions a_i, a_i', we must show:

$$\mathbb{E}[u_i(a_i, a_{-i}) \mid a_i] \;\geq\; \mathbb{E}[u_i(a_i', a_{-i}) \mid a_i].$$

Under π, the actions of other players are independent of a_i, so:

$$\mathbb{E}[u_i(a_i, a_{-i}) \mid a_i] = \mathbb{E}_{a_{-i} \sim \sigma_{-i}}[u_i(a_i, a_{-i})].$$

Similarly,

$$\mathbb{E}[u_i(a_i', a_{-i}) \mid a_i] = \mathbb{E}_{a_{-i} \sim \sigma_{-i}}[u_i(a_i', a_{-i})].$$

Since σ is a Nash equilibrium, a_i is a best response to σ_{-i} whenever it is played with positive probability. Hence,

$$\mathbb{E}_{a_{-i} \sim \sigma_{-i}}[u_i(a_i, a_{-i})] \;\geq\; \mathbb{E}_{a_{-i} \sim \sigma_{-i}}[u_i(a_i', a_{-i})].$$

Thus, π is a correlated equilibrium.

(2) Every Correlated Equilibrium is a Coarse Correlated Equilibrium

Let π be a correlated equilibrium. Then for every player i and every pair of actions a_i, a_i',

$$\mathbb{E}[u_i(a_i, a_{-i}) \mid a_i] \;\geq\; \mathbb{E}[u_i(a_i', a_{-i}) \mid a_i].$$

Multiplying both sides by $\Pr(a_i)$ and summing over all a_i, we get:

$$\sum_{a_i} \Pr(a_i)\, \mathbb{E}[u_i(a_i, a_{-i}) \mid a_i] \;\geq\; \sum_{a_i} \Pr(a_i)\, \mathbb{E}[u_i(a_i', a_{-i}) \mid a_i].$$

The left-hand side simplifies to:

$$\mathbb{E}_{a \sim \pi}[u_i(a)],$$

and the right-hand side becomes:

$$\mathbb{E}_{a \sim \pi}[u_i(a_i', a_{-i})].$$

Thus,

$$\mathbb{E}_{a \sim \pi}[u_i(a)] \;\geq\; \mathbb{E}_{a \sim \pi}[u_i(a_i', a_{-i})],$$

which is exactly the condition for a coarse correlated equilibrium.

Conclusion: Every Nash equilibrium induces a correlated equilibrium, and every correlated equilibrium is a coarse correlated equilibrium. Hence,

$$\text{Nash Equilibria} \subseteq \text{Correlated Equilibria} \subseteq \text{Coarse Correlated Equilibria}.$$

12. The definition of correlated equilibrium that we saw in class states that no player can benefit by unilaterally deviating from the coordinator's recommendation, where the deviation can be to any pure strategy. However, employing pure strategies is not the most general method of play. A player could in general mix over their pure strategies. The goal of this problem is to show that if players cannot improve their payoff by unilaterally deviating to any pure strategy, then they cannot improve their payoff by unilaterally deviating to any mixed strategy either. Consequently, the definition that we saw in class is general enough to encapsulate deviations to mixed strategies.

    Consider an $n-$player game and let $(\widehat{S}_1, \ldots, \widehat{S}_n) \sim p_{CE}$ where p_{CE} is a correlated equilibrium. Show that for every $i \in \{1, \ldots, n\}$, every $\widehat{s}_i \in \mathcal{S}_i$, and every $S_i' \sim q(\widehat{s}_i)$ where $q(\widehat{s}_i) \in \Delta(\mathcal{S}_i)$ is any mixed strategy,

$$\mathbb{E}[u_i(\widehat{S}_i, \widehat{S}_{-i}) | \widehat{S}_i = \widehat{s}_i] \geq \mathbb{E}[u_i(S_i', \widehat{S}_{-i}) | \widehat{S}_i = \widehat{s}_i].$$

    Solution:

    Since p_{CE} is a correlated equilibrium, for every $s_i' \in \mathcal{S}_i$,

$$\mathbb{E}[u_i(\widehat{S}_i, \widehat{S}_{-i}) \mid \widehat{S}_i = \widehat{s}_i] \geq \mathbb{E}[u_i(s_i', \widehat{S}_{-i}) \mid \widehat{S}_i = \widehat{s}_i].$$

    Let $S_i' \sim q(\widehat{s}_i)$. Then by linearity of expectation,

$$\mathbb{E}[u_i(S_i', \widehat{S}_{-i}) \mid \widehat{S}_i = \widehat{s}_i] = \sum_{s_i'} q(\widehat{s}_i)(s_i') \, \mathbb{E}[u_i(s_i', \widehat{S}_{-i}) \mid \widehat{S}_i = \widehat{s}_i].$$

    Applying the inequality above and summing, (note that probabilities are positive and sign of inequality remains intact)

$$\mathbb{E}[u_i(\widehat{S}_i, \widehat{S}_{-i}) \mid \widehat{S}_i = \widehat{s}_i] \geq \sum_{s_i'} q(\widehat{s}_i)(s_i') \, \mathbb{E}[u_i(s_i', \widehat{S}_{-i}) \mid \widehat{S}_i = \widehat{s}_i] = \mathbb{E}[u_i(S_i', \widehat{S}_{-i}) \mid \widehat{S}_i = \widehat{s}_i]$$

    Therefore,

$$\mathbb{E}[u_i(\widehat{S}_i, \widehat{S}_{-i}) \mid \widehat{S}_i = \widehat{s}_i] \geq \mathbb{E}[u_i(S_i', \widehat{S}_{-i}) \mid \widehat{S}_i = \widehat{s}_i].$$

    Hence proved.